2. [Maximum mark: 32]

Informally, the curvature of a curve can be thought of as the amount by which the curve deviates from being a straight line. In this question, you will investigate the curvature of a variety of functions.

Consider any function f that can be differentiated twice.

The curvature, k, of any function f is defined by $k(x) = \dfrac{|f''(x)|}{\left(1+\left(f'(x)\right)^2\right)^{\frac{3}{2}}}$.

Consider the family of linear functions $g(x) = mx + c$, where $x \in \mathbb{R}$ and $m, c \in \mathbb{R}$.

(a) Show that $k(x) = 0$ for this family of linear functions. [2]

Consider the family of quadratic functions $h(x) = ax^2 + bx + c$ for $x \in \mathbb{R}$, where $a \in \mathbb{R}$, $a \neq 0$ and $b, c \in \mathbb{R}$.

For this family of quadratic functions, it is given that

$$k(x) = \frac{2|a|}{\left(1+(2ax+b)^2\right)^{\frac{3}{2}}} \text{ and}$$

$$k'(x) = -\frac{12a|a|(2ax+b)}{\left(1+(2ax+b)^2\right)^{\frac{5}{2}}}.$$

The maximum value of $k(x)$ is denoted by $k_{\max}$.

(b) (i) By solving $k'(x) = 0$, find the value of x where $k_{\max}$ occurs.

You are **not** required to justify that this value of x leads to a maximum. [1]

(ii) Determine an expression for $k_{\max}$, in terms of a only. [2]

(iii) State the value of $\lim_{x\to\infty} k(x)$ and explain briefly the significance of this result. [2]

(iv) Consider the quadratic functions

$p(x) = -2x^2 + 2x - 10$ where $x \in \mathbb{R}$ and

$q(x) = 2x^2 + 5x + 25$ where $x \in \mathbb{R}$.

State which one of the following statements is true and justify your answer.

A. $k_{\max}$ of $p > k_{\max}$ of q

B. $k_{\max}$ of $p < k_{\max}$ of q

C. $k_{\max}$ of $p = k_{\max}$ of q [2]

(This question continues on the following page)

(Question 2 continued)

Consider the function $v(x) = \ln x$, where $x \in \mathbb{R}$, $x > 0$.

For v, it is given that

$$k(x) = \frac{x}{\left(1+x^2\right)^{\frac{3}{2}}} \text{ and}$$

$$k'(x) = \frac{1-2x^2}{\left(1+x^2\right)^{\frac{5}{2}}}.$$

(c) (i) Determine the exact value of x where $k_{\max}$ occurs.

You are **not** required to justify that this value of x leads to a maximum. [2]

(ii) Show that $k_{\max} = \dfrac{2\sqrt{3}}{9}$. [4]

Consider the function $w(x) = e^x$, where $x \in \mathbb{R}$.

For w, it is given that

$$k(x) = \frac{e^x}{\left(1+e^{2x}\right)^{\frac{3}{2}}} \text{ and}$$

$$k'(x) = \frac{e^x\left(1-2e^{2x}\right)}{\left(1+e^{2x}\right)^{\frac{5}{2}}}.$$

(d) (i) Show that $k_{\max} = \dfrac{2\sqrt{3}}{9}$. [5]

(ii) Suggest a reason why v and w have the same $k_{\max}$. [1]

Consider a family of curves $y = \sqrt{r^2 - x^2}$, where $-r < x < r$, $y > 0$ and r is a positive constant.

(e) (i) Show that $\dfrac{d^2y}{dx^2} = -\dfrac{r^2}{y^3}$. [6]

(ii) Hence, show that the curvature, k, is constant for this family of curves. [5]